

Complete increasing subsequences in multiset permutations (OEIS A047909 and A268485)

Quantitative Beta renewal asymptotics: central, diagonal and
separated-ratio expansions, and transition-uniform relative accuracy

Two AI-assisted research manuscripts, merged into one ProveIt report

Both Parts: 2 October 2026 merged: 2 October 2026

Abstract

A word chosen uniformly from all words with m copies of each letter $1, \dots, k$ contains $12 \cdots k$ with probability $p_m(k) = H(m, k)(m!)^k / (mk)!$, where $H(m, k)$ is the array A047909 and $H(n, n)$ is A268485. By a theorem of El Maazouz and Pitman, $p_m(k)$ is the probability that k independent $\text{Beta}(1, m)$ increments sum to at most 1, and $p_m(k)$ has a Gaussian limit in the window $k = m + O(\sqrt{m})$. This report prints two research manuscripts built on that representation. Part I proves an expansion to every fixed order in the central window, with the first four correction polynomials. It proves that the diagonal has only odd half-integer corrections beyond the leading $1/2$, and computes four of them. It also gives smooth probability and Lambert- W count inverses with qualified integer brackets, and proves an expansion to every fixed order of the exponentially small success or failure probability when k/m stays away from 1. Part II joins the two regimes. Unexpanded Gaussian–Hermite half-line kernels in the exponential-tilting construction give relative error $O(m^{-(R+1)/2})$ for both tails, uniformly on every fixed compact range of k/m in $(0, \infty)$, through the exact zero-saddle point $k = m + 1$. Part II recovers Part I’s first central and separated coefficients from this formula, and gives both real Lambert- W branches for separated tail thresholds. Effective constants, growing orders, ratios tending to 0 or ∞ and certified rounding remain open. The report is AI-assisted and unrefereed, and nothing in it is formalized.

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Guide to this report

The two Parts are two manuscripts of batch 77 (arrival commit 096ee7b87, 2 October 2026), placed in this directory by commit 4f11bc9c0. The second is an addendum to the first and cites it as its companion report:

Part	Batch-77 manuscript	Delivered title	Date
I	13	Quantitative Beta renewal asymptotics for complete increasing subsequences	2 Oct.
II	14	Transition uniform Beta renewal asymptotics for complete increasing subsequences	2 Oct.

Each Part prints its manuscript’s text, with every result, proof, remark, open question and limitation. Its statements keep their delivered wording. Part I keeps exactly the section, statement and equation numbers of manuscript 13 (Sections 1–10). Part II’s Section k is Section $k + 10$ here, so its equation $(k.n)$ is $(k + 10.n)$ and its Theorem 3.1 is Theorem 13.1. Each Part has its own symbols, fixed in its own text. Section “Notation across the two Parts” lists the symbols whose meaning changes between the Parts. Notes added when the manuscripts were merged are set in small type and tagged **[write, 2 October 2026]**. They give cross-references between the Parts, mark results that Part II re-derives, and mark the statements of Part I that Part II answers. No delivered statement was changed, and no symbol was renamed.

What is proved, and where

Statement	Where
Renewal identity $p_m(k) = \mathbb{P}(B_1 + \cdots + B_k \leq 1)$, $B_i \sim \text{Beta}(1, m)$, and the Gaussian limit	El Maazouz and Pitman [5], not new here. Direct proof of the identity: Part I, Proposition 2.1; restated in scaled form as Part II, (11.2)
Exact finite simplex formula	Part I, (2.2)
Central expansion to every fixed order; $P_1, \dots, P_4$	Part I, Theorem 3.1. Part II recovers P_1 ((16.2)) and identifies every P_j by uniqueness (Section 16)
Diagonal: only odd half-integer powers beyond $1/2$; four corrections	Part I, Corollary 4.1
Smooth probability inverse and integer brackets	Part I, Proposition 5.1
Lambert- W count inverse; strict increase of $a(n)$	Part I, Proposition 6.1 and the paragraph after it
Separated-ratio tails to every fixed order; c_1	Part I, Theorem 7.1. Part II recovers c_1 ((16.9)) and every c_j (Section 16)
Weighted $(1 + z^2)$ density remainder for the tilted triangular family	Part II, Lemma 14.1
Relative error $O(m^{-(R+1)/2})$ for both tails, uniformly for $a \leq k/m \leq b$; exact switch at $k = m + 1$	Part II, Theorem 13.1 and (12.4)
$p_m(m + 1) = \frac{1}{2} + \lambda_3 \phi(0)/(6\sqrt{k}) + O(k^{-3/2})$	Part II, (16.3)
Separated tail thresholds: both real Lambert- W branches, two inverse corrections, integer brackets	Part II, Corollary 17.1 and (17.7)
<i>Open after both Parts</i>	
Effective constants and certified integer thresholds; truncation order growing with m , coefficient growth, optimal truncation, Stokes phenomena or exponentially improved remainders; ratios k/m tending to 0 or ∞ ; growing windows for the central polynomial expansion (3.1) itself; a transition-uniform inversion; numerically stable evaluation of the exact probabilities and of the half-line kernels.	

Finite computations in both Parts (exact rational probabilities, word counts, symbolic re-derivations, quadrature at sample points) are checks and diagnostics; no asymptotic or uniform statement rests on them. The literature and repository searches recorded in the Parts are bounded, not novelty or priority certificates. Neither Part claims a new general Edgeworth, saddle-point or tilting method: Part II attributes the method to Esscher and Cramér through Daniels, with Lugannani–Rice, Robinson and Jensen as related classical work. The inverses use the Lambert W function. The leading count inverse of Part I is the case $a = b = 1$ of the exact Lambert-core theorem `p0:thm:lambert-core` of the repository’s transseries volume (`Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`), after taking logarithms. The correction terms of both Parts are instances of its perturbed inversion (`p0:thm:perturbed-inversion`). The integer brackets are of the first-crossing kind treated by its staircase recovery (`p0:thm:staircase`). No novelty is claimed for the inversion method.

Notation across the two Parts

Common to both Parts: $H(m, k)$, $W(m, k) = (mk)!/(m!)^k$, $p_m(k) = H(m, k)/W(m, k)$, $a(n) = H(n, n)$, the scaled increment $Y_m = mB$ with density f_m and transform M_m , the cumulant generating function $K_m(t) = \log \mathbb{E} e^{tY_m}$, $h = m^{-1/2}$, $x = (k - m)/\sqrt{m}$ (computed from the actual integer k), the exact ratio $\rho = k/m$, $I(\rho) = 1 - \rho + \rho \log \rho$, σ^2 and λ_r (tilted variance and standardized cumulants at the exact saddle), $\eta = k^{-1/2}$, ϕ , Φ , He_d , the central polynomials P_j and the separated coefficients

c_j . The symbols below change meaning between the Parts; each Part's own text fixes its meaning there.

Symbol	Part I	Part II
B	B_i : unscaled Beta(1, m) increments, success iff $\sum B_i \leq 1$; $B_1(t), B_2(t)$: transform corrections in (7.3)	Increments are the scaled $Y_{m,i}$, $S_k = \sum Y_{m,i}$, success iff $S_k \leq m$; B_1, B_2 : the same transform corrections; $B_R^e(u)$: the kernel sum of (12.6), not an increment
t	Fourier variable (Section 3); the exact saddle is t_m (Section 7)	The exact real saddle (Part I's t_m), of either sign or zero; Fourier variables are v, w
T	$T_{m,k}$: normalized sum in the proof of Theorem 3.1; T : frequency cutoff in (3.14); $T_m(k)$: the rare tail, defined only for ρ separated from 1	T : the outward saddle-side tail (13.1), defined for every ratio; on separated ranges it is Part I's $T_m(k)$
E, E_j	$E(h, u, x)$: the formal residual exponent (3.8); $E_j = [h^j]E$, polynomials in u, x	$E_j(z)$: $C_j(v)$ with each v^d replaced by $\text{He}_d(z)$. <i>Tempting false reading</i> : Part II's E_j is not Part I's $[h^j]E$
A	$A_j(u, x)$: coefficients of e^E (3.9); $A_*(n)$: smooth diagonal count carrier	$A = kK_m(t) - tm$: the tilted exponent; $A_0(\rho)$: the separated amplitude (16.6)
u	Formal variable standing for it (Section 3); integration variable in (7.7)	$u = t \sigma\sqrt{k} \geq 0$, the kernel argument
C, C_j	C : half-width of the central window $ x \leq C$	$C_j(v)$: coefficient polynomials (12.5); C : generic constants
Q	Q_R : polynomial in the bound (3.16); Q_j : exact-saddle coefficients (7.9)	$Q_R(w)$: the Gaussian–polynomial comparison (14.4); Q_1 in (16.5) is Part I's (7.11)
K	$K_m(t)$ as above; $K_{m,R}(p)$: root of the smooth probability carrier	$K_m(t)$ as above; $K_m(\ell)$: crossing of the separated carrier (Corollary 17.1). Same letter and subscript as the cumulant generating function; read it by its argument
F	$F_{m,R}(x)$: the smooth probability carrier	$F(u) = H_0(u) = e^{u^2/2}\Phi(-u)$, the scaled normal tail
H	$H(m, k)$ only	$H(m, k)$; also the half-line kernels $H_d(u)$ (12.6)
q	$q_m(p)$: integer threshold for a fixed probability p	$q < 1$: Cramér gap (14.2); q_s, q_f : integer thresholds for targets $e^{-m\ell}$ (17.6)
κ	$\kappa_r(m)$: cumulants of Y_m (3.7)	$\kappa = m\rho_0 + d_0 + d_1/m$: centre of the integer bracket, not a cumulant
L	$L = \log y$ (Proposition 6.1); a truncation order in (7.3)	A frequency cutoff (Section 14); $L = \log A_0$ (Corollary 17.1)
D, d	$D(n)$: diagonal probability carrier; $D_{R,\mathcal{P}}, D$: constants	$D_R(w)$: Fourier difference (14.4); d : a margin; d_0, d_1 : inverse coefficients (17.3)
J, a, b, g	J : fixed order; a, b : coefficients in the proof of Proposition 5.1; $a(n)$: A268485; $g_{m,k}$: tilted density	J : a compact saddle interval (Lemma 14.1) and a fixed separated order (Section 16); $0 < a < b$: ratio bounds; $g = \log \rho_0$
ε, ϵ	ε_m : rounding displacement; $\epsilon = +1$ for $\rho < 1$, -1 for $\rho > 1$	$\varepsilon = +1$ for $t \geq 0$ (that is, $k \leq m+1$), -1 otherwise; on separated ranges it equals Part I's ϵ
r	$r = \sqrt{2L/W_0(2L)}$	r : the density remainder

Part I's tilted-tail formula (7.10) expands the half-line weight $1/(t_m + iv)$ in powers of η , which requires $|t_m|$ bounded below. Part II keeps that weight unexpanded inside the kernels $H_d(u)$, which

are finite at $u = 0$. This difference is why Part I's separated theorem has poles at $\rho = 1$ and Part II's theorem has none.

Provenance and merge decisions

Two manuscripts, both of batch 77 (arrival commit 096ee7b87; placement commit 4f11bc9c0).

- **Part I**, manuscript 13, archive `beta-renewal-asymptotics-reproducibility.zip`, delivered main file `beta-renewal-asymptotics.tex` (14-page PDF), 2 October 2026, author line “Research report for OEIS A047909 and A268485”. Pin 4b874cea0, the commit of its scoped repository overlap check (Section 9). Contributes the central, diagonal, inverse and separated-ratio theories, its proof notes and seven check scripts.
- **Part II**, manuscript 14, archive `beta-renewal-uniform-reproducibility.zip`, delivered main file `beta-renewal-uniform-addendum.tex` (12-page PDF), 2 October 2026, author line “Research addendum for OEIS A047909 and A268485”. It names no repository commit. Its `source-hashes.json` pins manuscript 13 instead: the SHA-256 values of 13's TeX, PDF and archive equal the delivered bytes. Contributes the weighted density lemma, the transition-uniform theorem, the recovery of P_1 and c_1 , the separated Lambert-branch inverse, and three check scripts.

Where the merge had to choose:

1. *Order and base*. Manuscript 14 is an addendum to manuscript 13 and says that it does not replace it. Manuscript 13 is the base and Part I; nothing is superseded.
2. *Re-derived material*. Part II restates Part I's definitions, the renewal identity in scaled form, the transform expansion (7.3) and the saddle displacement (7.6), and recomputes Q_1 , the three contributions to c_1 , c_1 itself and P_1 . These are printed where they stand, because Part II's proofs cite them. `[write]` notes identify them with Part I's equations. Part II's recovery of P_1 and c_1 is printed as a consistency check by a second route, not as a second statement of Part I's theorems.
3. *Answered non-claims*. Part I says that its two regimes are proved separately and that no transition-uniform matching is claimed, and lists the uniform bridge as its first open question. These passages are printed as delivered, each followed by a note pointing to Part II. The questions that Part II answers only in part are re-scoped in the notes.
4. *References between manuscripts*. Manuscript 14 cites manuscript 13 as [6]. The citation is kept, and the bibliography entry says that the companion report is Part I.
5. *Front matter*. The two title blocks and abstracts are printed at the head of their Parts; manuscript 13's table of contents is replaced by this report's.
6. *Bibliography*. The two bibliographies are merged into one. Five entries occur in both. Manuscript 13's copies of the two OEIS entries add an inspection date and an indexing note; those copies are kept, and the other three shared entries are identical in content.
7. *Preamble*. One preamble, the union of the two; every shared macro has the same definition in both sources. The `\pdfmapfile` lines of both preambles were removed; on MiKTeX they only produce “fontmap entry already exists” warnings. Manuscript 13 set no paper size and

manuscript 14 set letter paper; this report uses letter paper throughout. The running heads of the two manuscripts are replaced by one.

8. *Labels*. Every label carries the report prefix: **brs:** in Part I and **brs:un:** in Part II, followed by the delivered name. Eight delivered names occur in both manuscripts, for different equations (**eq:renewal**, **eq:density**, **eq:Kexp**, **eq:tshift**, **eq:rare**, **eq:c1**, **eq:Q1**, **eq:ceil**).

Part I

Quantitative Beta renewal asymptotics for complete increasing subsequences

Batch-77 manuscript 13, archive `beta-renewal-asymptotics-reproducibility.zip` (delivered as `beta-renewal-asymptotics.tex`, 14 pages), dated 2 October 2026. Delivered author line: “Research report for OEIS A047909 and A268485”. Pin 4b874cea0. Printed as delivered, with **[write]** notes; section, statement and equation numbers are those of the delivery.

Abstract of this Part (as delivered). A word chosen uniformly from all words with m copies of each letter $1, \dots, k$ contains $12 \cdots k$ with probability $p_m(k)$. The exact representation of this probability as the tail of a renewal process with independent $\text{Beta}(1, m)$ increments, and its Gaussian limit, are established results of El Maazouz and Pitman. Starting from that representation, we give an explicit expansion to every fixed finite order in the central window $k = m + O(\sqrt{m})$, prove its uniform remainder, and compute the first four central polynomials. On the diagonal, a parity argument eliminates all integer powers beyond the leading $1/2$ and gives four explicit half-integer corrections. We derive smooth success thresholds and a Lambert- W inverse for diagonal counts, keeping the integer rounding issue explicit. A complementary saddle calculation gives the exponentially small success or failure probability, to every fixed integer correction order, when k/m stays in a compact set separated from 1. The two regimes are proved separately; no transition-uniform matching or optimal truncation is claimed. Exact rational checks and independent symbolic and combinatorial calculations accompany the report.

[write, 2 October 2026] The sentence “The two regimes are proved separately; no transition-uniform matching or optimal truncation is claimed” is true of this Part. Part II proves transition-uniform matching: Theorem 13.1 gives one formula with relative error $O(m^{-(R+1)/2})$ for both tails, uniformly on every fixed compact range of k/m in $(0, \infty)$, and Section 16 re-expands it into this Part’s P_j and c_j . Optimal truncation is claimed in neither Part.

1 The counting problem and the scope of the report

Let $H(m, k)$ be the number of words with exactly m copies of each symbol in $\{1, \dots, k\}$ that contain the subsequence $12 \cdots k$. A subsequence need not be contiguous. Because there are only k distinct letters, this condition is equivalent to having a strictly increasing subsequence of length k . Define

$$W(m, k) = \frac{(mk)!}{(m!)^k}, \quad p_m(k) = \frac{H(m, k)}{W(m, k)}, \quad m, k \geq 1. \quad (1.1)$$

Thus $p_m(k)$ is a probability rather than a raw count.

OEIS indexing. The array A047909 is $h(m, k) = H(m, k)$, with m the row index (multiplicity) and k the column index (alphabet size). Its row $m = 2$ is 1, 5, 47, 641, 11389, $\dots$, whereas column $k = 2$ is 1, 5, 19, 69, 251, $\dots$. The flattened sequence is read by upward antidiagonals: antidiagonal $d \geq 1$ is

$$H(d, 1), H(d-1, 2), \dots, H(1, d).$$

The diagonal sequence A268485 is

$$a(n) = H(n, n) \quad (n \geq 1), \quad a(0) = 1. \quad (1.2)$$

The $n = 0$ convention is the empty word and is separate from the Beta construction below. These conventions agree with the official OEIS entries [1, 2].

Established results. Horton and Kurn [3] supplied exact enumeration. Their formula is reproduced in Clifton, Deb, Huang, Spiro and Yoo [4], which also studies means and states Gaussian and moment conjectures. El Maazouz and Pitman [5] subsequently established the Beta increment mechanism and the Gaussian limit, in particular in their Proposition 5.13. The latter paper was published online in November 2023 and in volume 33 of *Combinatorics, Probability and Computing* in 2024. Accordingly, neither the renewal identity nor the Gaussian limit is a newly resolved conjecture in this report.

Quantitative contribution and its limits. The work here is the explicit specialization to this counting family: a uniform central expansion at every fixed finite order, its coefficient algorithm, the diagonal parity and corrections, controlled inversions, and a complementary separated-ratio rare-tail expansion. A bounded literature and repository search did not locate these particular explicit higher-order central and inverse formulas. This is not a universal historical novelty certificate, and no new general Edgeworth or saddle method is claimed. All asymptotic orders are fixed before the limit; constants may depend on that order and on the stated compact set. Section 9 gives the source-audit boundaries.

2 The exact Beta renewal law

Proposition 2.1 (Established identity with a direct proof). *Let $B_1, B_2, \dots$ be independent with density $m(1-b)^{m-1}$ on $0 < b < 1$. Then*

$$p_m(k) = \mathbb{P}(B_1 + \dots + B_k \leq 1). \quad (2.1)$$

Proof. Give each of the mk distinguishable cards an independent uniform position in $(0, 1)$. Sort the positions and forget copy labels. Each multiset word has exactly $(m!)^k$ distinguishable preimages, so the resulting word is uniform. Read greedily: after selecting the earliest possible 1, choose the earliest 2 after it, and continue. Greedy success is equivalent to the existence of $12 \dots k$.

Suppose previous greedy selections have ended at position t . The m positions belonging to the next letter have not been inspected; they remain independent uniforms and are independent of all earlier letters. For $0 \leq w \leq 1 - t$, the probability of no such position in $(t, t + w]$ is $(1 - w)^m$. Consequently the next successful wait has density $m(1 - w)^{m-1}$ on $(0, 1 - t)$, and the failure atom has mass t^m . This is exactly the kernel obtained by drawing an independent Beta(1, m) increment and killing the process if it exceeds $1 - t$, since $\mathbb{P}(B > 1 - t) = t^m$. The entire killed kernels agree, including their failure masses. Induction proves (2.1). $\square$

It is important that the proof does not replace a wait conditioned on success by an unconditioned increment. The killing construction is what makes independence valid. If

$$N_m(1) = \max\{j \geq 0 : B_1 + \dots + B_j \leq 1\},$$

then $\mathbb{P}(N_m(1) \geq k) = p_m(k)$ and $N_m(1) \geq 1$ almost surely. This fixes the stopping-index convention directly; shifted conventions in related formulations do not change any formula here.

2.1 An exact finite arithmetic formula

On the simplex $b_i \geq 0$, $\sum b_i \leq 1$, expand the product of the Beta densities and use the Dirichlet monomial integral. This gives

$$p_m(k) = m^k \sum_{j_1, \dots, j_k=0}^{m-1} \frac{\prod_{i=1}^k (-1)^{j_i} \frac{(m-1)!}{(m-1-j_i)!}}{(k+j_1+\dots+j_k)!}. \quad (2.2)$$

For computation, take the k th power of $\sum_{j=0}^{m-1} (-1)^j (m-1)!/(m-1-j)! z^j$, and perform the remaining sum over its coefficients. The common denominator $(mk)!$ makes the calculation exact using integers and rational numbers. Direct floating point evaluation of the alternating sum is inappropriate because of severe cancellation. This formula supplies finite checks independently of the asymptotic derivation.

3 A uniform central expansion at every fixed order

Write

$$h = m^{-1/2}, \quad x = \frac{k-m}{\sqrt{m}}, \quad \phi(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}, \quad \Phi(x) = \int_{-\infty}^x \phi(u) du.$$

Theorem 3.1 (Central expansion). *For every fixed integer $R \geq 0$ and fixed $C < \infty$, there are explicitly computable polynomials $P_1, \dots, P_R$ such that, uniformly over integers m, k with $m \rightarrow \infty$ and $|x| \leq C$,*

$$p_m(k) = \Phi(-x) + \phi(x) \sum_{j=1}^R P_j(x) m^{-j/2} + O_{R,C}(m^{-(R+1)/2}). \quad (3.1)$$

The first four polynomials are

$$P_1(x) = \frac{x^2 + 8}{6}, \quad (3.2)$$

$$P_2(x) = \frac{x(x^4 + 11x^2 - 6)}{72}, \quad (3.3)$$

$$P_3(x) = \frac{5x^8 + 40x^6 - 561x^4 - 1074x^2 - 4848}{6480}, \quad (3.4)$$

$$P_4(x) = \frac{x(5x^{10} - 5x^8 - 1539x^6 + 3339x^4 + 15264x^2 + 173988)}{155520}. \quad (3.5)$$

For $R = 0$, this is the known Gaussian transition together with an $O_C(m^{-1/2})$ error. The theorem gives each requested finite refinement; it asserts neither convergence of an infinite formal series nor uniformity when R or C grows with m .

3.1 Finite coefficient construction

Set $Y_m = mB$. Its moments and cumulants satisfy

$$\mu_0(m) = 1, \quad \mu_r(m) = r! \prod_{a=1}^r (1 + a/m)^{-1}, \quad (3.6)$$

$$\kappa_r(m) = \mu_r(m) - \sum_{a=1}^{r-1} \binom{r-1}{a-1} \kappa_a(m) \mu_{r-a}(m). \quad (3.7)$$

Only $r \leq R + 2$ is needed for order R . With u formal, form through degree h^R the expression

$$E(h, u, x) = (h^{-1} + x)(\kappa_1 - 1)u + \frac{(1 + xh)\kappa_2 - 1}{2}u^2 \\ + (1 + xh) \sum_{r=3}^{R+2} h^{r-2} \kappa_r \frac{u^r}{r!}. \quad (3.8)$$

Here and below each cumulant is expanded to the finite required order in h^2 . If $E_j = [h^j]E$, define

$$A_0 = 1, \quad A_j(u, x) = \frac{1}{j} \sum_{i=1}^j i E_i(u, x) A_{j-i}(u, x). \quad (3.9)$$

These are the coefficients of the finite exponential e^E . Using the probabilists' Hermite polynomials $\text{He}_d(z) = (-1)^d e^{z^2/2} (d/dz)^d e^{-z^2/2}$, the answer is

$$P_j(x) = - \sum_{d \geq 1} [u^d] A_j(u, x) \text{He}_{d-1}(-x). \quad (3.10)$$

All sums and series manipulations in this prescription are finite. In particular, this procedure does not assume analyticity in h of the full moment generating function of the Beta distribution.

3.2 Uniform analytic estimates

We give the details that turn the formal prescription into a theorem. For $m \geq 2$, the density of Y_m is

$$f_m(y) = (1 - y/m)^{m-1} \mathbf{1}_{(0,m)}(y) \leq e^{-y/2}. \quad (3.11)$$

Thus $M_m(z) = \mathbb{E} e^{zY_m}$ has uniformly bounded exponential moments on a fixed complex neighborhood of zero. By $|e^{zy} - 1| \leq |z|ye^{|z|y}$, one can choose a fixed disk on which $|M_m(z) - 1| < 1/2$. A single analytic branch of $\log M_m(z)$ exists there, and its derivatives of any prescribed fixed order are uniformly bounded. The variance is

$$\kappa_2(m) = \frac{m^3}{(m+1)^2(m+2)} \rightarrow 1.$$

Taylor's theorem therefore gives fixed $\delta, c > 0$ such that

$$|\mathbb{E} e^{itY_m}| \leq e^{-ct^2}, \quad |t| \leq \delta, \quad (3.12)$$

for all sufficiently large m .

Dominated convergence in (3.11) gives L^1 convergence of f_m to the unit exponential density. Their characteristic functions consequently converge uniformly on the entire real line to $(1 - it)^{-1}$. For any fixed $T > \delta$, this implies a uniform $\eta_0 < 1$ bound on the annulus $\delta \leq |t| \leq T$. The density f_m is decreasing from 1 to 0; integration by parts, including the boundary at zero, gives

$$|\mathbb{E} e^{itY_m}| \leq \frac{2}{|t|}. \quad (3.13)$$

For $T > 4$ one obtains, on the positive half line,

$$\int_{\delta}^{\infty} |\mathbb{E} e^{itY_m}|^k \frac{dt}{t} \leq \eta_0^k \log(T/\delta) + \frac{(2/T)^k}{k}. \quad (3.14)$$

The constants can be chosen independently of all sufficiently large m . These estimates supply the small-, middle-, and high-frequency controls needed for Fourier inversion in the triangular array.

3.3 Proof of the central remainder

Proof of Theorem 3.1. Define

$$T_{m,k} = \frac{\sum_{i=1}^k Y_{m,i} - k}{\sqrt{m}}.$$

For $|th|$ inside the fixed nonvanishing disk, choose the logarithm branch continuous from $t = 0$. The characteristic logarithm is then exactly

$$\log \mathbb{E} e^{itT_{m,k}} = k\{\log M_m(ith) - ith\}. \quad (3.15)$$

Since $k = h^{-2} + xh^{-1}$, the leading term is $-t^2/2$ and the residual formal exponent is (3.8) at $u = it$. Taylor expansion of the logarithm through degree $R + 2$ in its argument leaves an error $O(h^{R+1}|t|^{R+3})$, uniformly in bounded x . Each of the finitely many rational cumulants in (3.7) has a finite Taylor expansion in h^2 with a controlled remainder.

Choose a fixed $\alpha > 0$ with $\alpha < 1/[6(R + 3)]$. On $|t| \leq m^\alpha$, exponentiate the finite correction through degree R . Finite Taylor remainder estimates, with a slightly weakened Gaussian majorant from (3.12), bound the difference from

$$e^{-t^2/2} \sum_{j=0}^R h^j A_j(it, x)$$

by

$$C_{R,C} h^{R+1} Q_R(|t|) e^{-c't^2}, \quad (3.16)$$

where Q_R is a fixed polynomial with zero constant term. The cutoff is small enough that the exponent correction and its finite remainders are uniformly controlled. Polynomial factors are absorbed by the weakened Gaussian.

Between m^α and $\delta\sqrt{m}$, the Gaussian estimate makes the inversion integrals smaller than every power of m^{-1} . Beyond $\delta\sqrt{m}$, substitute the unscaled frequency th and apply (3.14); because $k/m \rightarrow 1$ uniformly in the central window, this contribution is exponentially small. The normal-polynomial comparison has the same required integrable tails.

The exact sum and the comparison function have continuous integrable densities for all sufficiently large k ; the latter need not be positive. Their characteristic functions agree at zero. The difference of their CDFs is bounded by the integral of their characteristic-function difference divided by $|t|$, up to $1/(2\pi)$. The low-frequency bound (3.16) is integrable after this division because $Q_R(0) = 0$. The other frequency ranges were just controlled. This gives an $O(h^{R+1})$ CDF error, uniformly in the threshold.

The inverse transform of $(it)^d e^{-t^2/2}$ is $\phi(z) \text{He}_d(z)$, whose integral from $-\infty$ to z is $-\phi(z) \text{He}_{d-1}(z)$ for $d \geq 1$. Finally, $\sum B_i \leq 1$ is equivalent to $T_{m,k} \leq -x$. Evaluating at $z = -x$ yields (3.10) and (3.1). $\square$

4 The diagonal expansion and its parity

Corollary 4.1 (Diagonal A268485). *As $n \rightarrow \infty$,*

$$\begin{aligned} \frac{a(n)}{(n^2)!/(n!)^n} = \frac{1}{2} + \frac{1}{\sqrt{2\pi}} \Big\{ \frac{4}{3n^{1/2}} - \frac{101}{135n^{3/2}} + \frac{19819}{7560n^{5/2}} \\ - \frac{4463177}{272160n^{7/2}} + O(n^{-9/2}) \Big\}. \end{aligned} \quad (4.1)$$

More generally, only odd half-integer powers occur beyond $1/2$. Retaining through $n^{-(2J+1)/2}$, for any fixed $J \geq 0$, leaves a remainder $O_J(n^{-(2J+3)/2})$.

Proof. Set $x = 0$ in (3.8). Every monomial $h^j u^d$ has $j \equiv d \pmod{2}$, because every cumulant is a series in h^2 . This property is preserved by products and therefore by the finite exponential. In (3.10), $\text{He}_{d-1}(0) = 0$ for even d . Thus only odd j survives. For the stated remainder, apply Theorem 3.1 at $R = 2J + 2$, whose last even coefficient vanishes. Finite rational calculation gives the four displayed coefficients. $\square$

The leading half-factor is consequently refined, rather than newly identified. For example, exact arithmetic gives

$$\sqrt{40}\{p_{40}(40) - \tfrac{1}{2}\} = 0.5250319715431282\dots, \quad \frac{4}{3\sqrt{2\pi}} = 0.5319230405352436\dots$$

The approach of the first number to the second is consistent with the negative $n^{-3/2}$ correction. Finite numerical agreement is a useful sign and indexing check; the Fourier proof supplies the asymptotic error statement.

5 Inverting success probabilities

Fix a compact interval $\mathcal{P} \subset (0, 1)$ and put $z_p = \Phi^{-1}(p)$. To avoid inventing an unstated interpolation of an integer-indexed probability, define a specific smooth carrier

$$F_{m,R}(x) = \Phi(-x) + \phi(x) \sum_{j=1}^R P_j(x) m^{-j/2}.$$

On a fixed central interval containing the required normal quantiles, $F'_{m,R}(x) = -\phi(x) + O(m^{-1/2})$ is uniformly negative for large m . Let $K_{m,R}(p) = m + x\sqrt{m}$, where x is the unique root in this interval of $F_{m,R}(x) = p$. This is a chosen approximation, not a canonical extension of $p_m(k)$ to real k .

Proposition 5.1 (Smooth inverse and integer bracket). *For each fixed $R \geq 2$, uniformly for $p \in \mathcal{P}$,*

$$K_{m,R}(p) = m - z_p \sqrt{m} + \frac{z_p^2 + 8}{6} + \frac{z_p(z_p^2 + 38)}{72\sqrt{m}} + O(m^{-1}). \quad (5.1)$$

Define the exact integer threshold by

$$q_m(p) = \min\{k \geq 1 : p_m(k) \leq p\}.$$

For a suitable uniform constant $D_{R,\mathcal{P}}$ and all sufficiently large m , with $\varepsilon_m = D_{R,\mathcal{P}} m^{-R/2}$,

$$\lceil K_{m,R}(p) - \varepsilon_m \rceil \leq q_m(p) \leq \lceil K_{m,R}(p) + \varepsilon_m \rceil. \quad (5.2)$$

Proof. Insert $x = -z + ah + bh^2$ into $F_{m,R}(x) = \Phi(z)$ and compare coefficients. At order h , $a = P_1(-z) = (z^2 + 8)/6$. At order h^2 ,

$$b = -\frac{za^2}{2} + a\{P'_1(-z) + zP_1(-z)\} + P_2(-z) = \frac{z(z^2 + 38)}{72}.$$

The implicit derivative is bounded away from zero uniformly for $p \in \mathcal{P}$, so the finite Taylor remainder controls the inverse remainder. Higher fixed orders follow by the same recursion.

The increments are positive, with positive density on $(0, 1)$, so $p_m(k)$ is strictly decreasing in k and tends to zero. The threshold exists. The probability error in Theorem 3.1 is $O(h^{R+1})$; the carrier derivative with respect to k has magnitude at least ch on the relevant interval. A displacement Dh^R therefore absorbs the probability error when D is sufficiently large. Integers below $K_{m,R} - Dh^R$ have probability greater than p , and integers at or above $K_{m,R} + Dh^R$ have probability at most p . Taking ceilings gives (5.2). $\square$

If an asymptotic truncation replaces the exact carrier root, enlarge ε_m to include its root error. In particular, (5.1) alone has smooth-root error $O(m^{-1})$. When the two ceilings in (5.2) agree, they determine the exact integer threshold. Near an integer boundary, sharper error control or exact evaluation is needed. There is no vanishing-error expansion of an integer threshold obtained by simply suppressing its lattice oscillation.

6 Inverting diagonal count growth

For large real n , choose a finite diagonal probability carrier $D(n)$ from Corollary 4.1 and set

$$A_*(n) = \frac{\Gamma(n^2 + 1)}{\Gamma(n + 1)^n} D(n).$$

For sufficiently large n this is positive and increasing. Different finite carriers agree to the order for which their probability expansions are retained; no unique real interpolation of $a(n)$ is asserted. Stirling subtraction gives

$$\begin{aligned} \log A_*(n) &= n^2 \log n - \frac{n}{2} \log(2\pi n) + \log n \\ &\quad + \frac{1}{2} \log(2\pi) - \frac{1}{12} - \log 2 + O(n^{-1/2}). \end{aligned} \tag{6.1}$$

Indeed the factorial-ratio logarithm alone has the displayed terms without $-\log 2$ and with error $O(n^{-2})$; the probability contributes $-\log 2 + O(n^{-1/2})$.

Proposition 6.1 (A Lambert- W count inverse). *Put $L = \log y$ and*

$$r = \sqrt{\frac{2L}{W_0(2L)}},$$

where W_0 is the positive real branch of Lambert's W function. If $A_(n_*(y)) = y$, then*

$$n_*(y) = r + \frac{\log(2\pi r)}{4 \log r + 2} + O(r^{-1}). \tag{6.2}$$

In particular, the smooth displacement $n_(y) - r$ tends to $1/4$.*

Proof. The definition gives $r^2 \log r = L$. The leading logarithmic derivative is $r(2 \log r + 1)$. One Newton correction for the next term in (6.1) is therefore $\log(2\pi r)/(4 \log r + 2)$. Substitution at this corrected point leaves a logarithmic residual $O(\log r)$: the quadratic leading-term remainder, the change in the subleading term, and the uncorrected $\log r$ term all have at most that size. Dividing by a derivative asymptotic to $r(2 \log r + 1)$ gives the $O(r^{-1})$ inverse error. $\square$

Finite Stirling and diagonal expansions, followed by Newton or formal implicit inversion, give arbitrary further fixed terms. At any specified order a logarithmic error is converted into a smooth inverse error by division by a quantity asymptotic to $n(2\log n + 1)$.

[write, 2 October 2026] The leading inverse r solves $r^2 \log r = L$, that is $X \log X = 2L$ with $X = r^2$. After taking logarithms, $\log X + \log \log X = \log(2L)$ is the case $a = b = 1$ of the exact Lambert-core theorem `p0:thm:lambert-core` of the repository's transseries volume (`Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`). The Newton correction in (6.2), like the inverse expansion (5.1), is an instance of its perturbed inversion `p0:thm:perturbed-inversion`. The integer brackets (5.2) and the first-crossing discussion below are of the kind treated by its staircase recovery `p0:thm:staircase`. No novelty is claimed for the inversion method.

For clarity, exact diagonal counts are strictly increasing for $n \geq 1$. An injection from successful n -square words appends the block $1, \dots, n$, followed by $n + 1$ copies of the new largest letter. The result is a successful $(n + 1)$ -square word. Successful target words beginning with the new largest letter lie outside the image. This proves strictness and makes a first-crossing index well-defined. Nevertheless, integer crossings must be bracketed using controlled errors. The limit $1/4$ in Proposition 6.1 concerns the smooth carrier; it is not a claim about an integer displacement.

7 Exponentially small tails away from the transition

Now let $\rho = k/m$ be the *exact* integer ratio. Throughout this section, ρ lies in a fixed compact set $\mathcal{K}$ contained either in $(0, 1)$ or in $(1, \infty)$. Define

$$I(\rho) = 1 - \rho + \rho \log \rho, \quad T_m(k) = \begin{cases} 1 - p_m(k), & \rho < 1, \\ p_m(k), & \rho > 1. \end{cases}$$

The first case is rare failure and the second rare success.

Theorem 7.1 (Separated-ratio expansion). *For every fixed integer $J \geq 0$, uniformly on $\mathcal{K}$,*

$$T_m(k) = \frac{e^{1-1/\rho} \sqrt{\rho}}{|\rho - 1| \sqrt{2\pi m}} e^{-mI(\rho)} \times \left\{ 1 + \sum_{j=1}^J \frac{c_j(\rho)}{m^j} + O_{J,\mathcal{K}}(m^{-J-1}) \right\}, \quad (7.1)$$

where the c_j are explicitly computable rational functions, possibly singular at 0 and 1. In particular,

$$c_1(\rho) = -\frac{\rho^4 + 10\rho^3 - 17\rho^2 + 24\rho - 6}{12\rho^3(\rho - 1)^2}. \quad (7.2)$$

Replacing k/m by its limit in (7.1) without controlling the offset changes the prefactor. Even a bounded displacement in k can produce a nonvanishing exponential factor. Likewise the poles at $\rho = 1$ prevent the theorem from being applied uniformly as the ratio approaches the central window.

[write, 2 October 2026] This remains true of Theorem 7.1. Part II keeps the half-line weight of (7.8) unexpanded inside the kernels $H_d(u)$ of (12.6), which are finite at $u = 0$. Its Theorem 13.1 is uniform through $\rho = 1$ and through the exact zero-saddle point $k = m + 1$, and (16.9) recovers the coefficient (7.2) from it.

7.1 Finite transform and saddle expansions

For real $t < 1$, set $K_m(t) = \log \mathbb{E} e^{tY_m}$. On compact real intervals below 1, and on a fixed sufficiently small complex neighborhood, finite density expansion gives, with derivative-uniform remainders,

$$\begin{aligned} K_m(t) &= -\log(1-t) + \sum_{j=1}^L m^{-j} B_j(t) + O_L(m^{-L-1}), \\ B_1(t) &= \frac{1}{1-t} - \frac{1}{(1-t)^2}, \\ B_2(t) &= \frac{3}{2(1-t)^2} - \frac{4}{(1-t)^3} + \frac{5}{2(1-t)^4}. \end{aligned} \quad (7.3)$$

To justify every finite order, start from

$$(m-1) \log(1-y/m) = -y + \sum_{j \geq 1} m^{-j} \left(\frac{y^j}{j} - \frac{y^{j+1}}{j+1} \right). \quad (7.4)$$

Split the integral at $y = m^\alpha$ for a sufficiently small fixed α . On the first part, expand to the needed finite order and bound the remainder by a polynomial times a common integrable exponential. On the second part, exponential domination applies uniformly in the complex neighborhood, whose real part stays below 1 by a fixed margin. Polynomial factors also control every prescribed finite derivative. Integrating the finite polynomial coefficients against $e^{-(1-t)y}$ and taking a finite formal logarithm gives the rational functions B_j . Analytic dependence of the full transform on $1/m$ is not required.

There is a unique real saddle t_m with

$$K'_m(t_m) = \rho^{-1}. \quad (7.5)$$

Uniform strict convexity near the limiting saddle and (7.3) yield

$$t_m = 1 - \rho + \frac{2/\rho - 1}{m} + O(m^{-2}), \quad K''_m(t_m) = \rho^{-2} + O(m^{-1}). \quad (7.6)$$

Successive substitution in (7.5), with a finite-order implicit-function estimate for the residual, gives all further fixed orders. The derivative at the limiting saddle is ρ^{-2} and is uniformly nonzero on $\mathcal{K}$.

7.2 Exact tilting and the integer-power algorithm

Tilt the density to $e^{t_m y} f_m(y)/M_m(t_m)$. Write $\sigma^2 = K''_m(t_m)$ and $\lambda_r = K_m^{(r)}(t_m)/\sigma^r$. The tilted mean is $1/\rho$. If $g_{m,k}$ is the tilted density of $\sum_{i=1}^k Y_{m,i} - m$, then, with $\epsilon = +1$ for $\rho < 1$ and $\epsilon = -1$ for $\rho > 1$,

$$T_m(k) = e^{kK_m(t_m) - t_m m} \int_0^\infty e^{-|t_m|u} g_{m,k}(\epsilon u) du. \quad (7.7)$$

Equivalently, Fourier inversion and integration of the half-line weight give

$$\begin{aligned} T_m(k) &= \frac{e^{kK_m(t_m) - t_m m}}{2\pi} \int_{\mathbb{R}} \left\{ \frac{M_m(t_m + iv)}{M_m(t_m)} e^{-iv/\rho} \right\}^k \\ &\quad \times \frac{\operatorname{sgn}(t_m)}{t_m + iv} dv. \end{aligned} \quad (7.8)$$

The global formula uses the characteristic function directly; the analytic branch K_m is used only near the real saddle, where the transform is nonvanishing. There is no real-axis singularity: $|t_m|$ is uniformly bounded away from zero. The interchange is justified by the integrable k th power of the tilted characteristic function and the integrable half-line weight.

Let $\eta = k^{-1/2}$ and let $\mathcal{G}$ be the linear Gaussian functional on polynomials in u defined by

$$\mathcal{G}[u^{2a}] = (-1)^a (2a - 1)!!, \quad \mathcal{G}[u^{2a+1}] = 0, \quad \mathcal{G}[1] = 1.$$

The negative signs encode powers of the imaginary Fourier argument. For $j \geq 0$, set

$$Q_j = \mathcal{G} \left([\eta^{2j}] \left\{ \frac{1}{1 + \eta u / (t_m \sigma)} \exp \left(\sum_{r=3}^{2j+2} \lambda_r \frac{u^r \eta^{r-2}}{r!} \right) \right\} \right). \quad (7.9)$$

Then $Q_0 = 1$, and the exact-saddle expansion is

$$T_m(k) = \frac{e^{kK_m(t_m) - t_m m}}{|t_m| \sqrt{2\pi k K_m''(t_m)}} \left\{ \sum_{j=0}^J Q_j k^{-j} + O_{J,K}(k^{-J-1}) \right\}. \quad (7.10)$$

The Q_j here use the actual m -dependent tilted cumulants and saddle; they are not yet the $c_j(\rho)$ in (7.1). Substitute the finite transform and saddle expansions into (7.10), use $k = \rho m$, and collect powers of m^{-1} to obtain those c_j . This is a finite algorithm at any prescribed order.

7.3 Uniform remainder and the first correction

Proof of Theorem 7.1. The tilted laws have uniformly bounded exponential moments and variances bounded above and below. Their densities converge uniformly in the parameter, in L^1 , to exponential densities of rate ρ . The annular nonlattice bound therefore follows as in the central proof. For large m , their logarithmic density derivative is

$$t_m - \frac{m-1}{m-y} < 0,$$

since $t_m \leq 1 - a$ for a fixed $a > 0$. The endpoint density and total variation are uniformly bounded, yielding a uniform $O(1/|v|)$ characteristic-function bound. These facts give the same three frequency controls used in Theorem 3.1.

In (7.8), scale $v = w/(\sigma\sqrt{k})$ and expand the exponent and denominator to any fixed order on a small-power cutoff. The leading factor is $e^{-w^2/2}$, and all remainders are bounded by a fixed polynomial times a weaker integrable Gaussian. Odd powers of η integrate to zero: each of their monomials has odd degree in the imaginary Fourier variable. Expand through one additional odd order before bounding the remainder; after retaining k^{-J} this gives $O(k^{-J-1})$. Middle and high frequencies are smaller than every algebraic order. The coefficients are exactly (7.9), proving (7.10) with its uniform error.

Stationarity at $t_0 = 1 - \rho$ gives

$$kK_m(t_m) - t_m m = -mI(\rho) + 1 - \rho^{-1} + O(m^{-1}).$$

Together with (7.6), this gives the leading factor of (7.1). Repeating the finite substitutions proves all stated orders and the rational form of the coefficients. For the first order,

$$Q_1 = \frac{\lambda_4}{8} - \frac{5\lambda_3^2}{24} - \frac{\lambda_3}{2t_m\sigma} - \frac{1}{t_m^2\sigma^2}. \quad (7.11)$$

The order- m^{-1} contributions from the exponent, the prefactor, and Q_1/k , respectively, are

$$\frac{2\rho^2 - 4\rho + 1}{2\rho^3}, \quad -\frac{\rho^2 - 3\rho + 1}{\rho^2(\rho - 1)}, \quad \frac{1}{\rho} \left(-\frac{1}{12} - \frac{\rho}{(1 - \rho)^2} \right).$$

Their sum is (7.2). In particular the true saddle displacement has been included; omitting it changes the correction. $\square$

At $\rho = 2$, $m = 40$, exact arithmetic gives a ratio of exact tail to leading approximation of about 0.98318167. Including c_1/m changes the ratio of exact tail to the approximation to about 1.00143703. This illustrates the correction, not an effective universal error bound. Convergence can be much slower near $\rho = 1$, where the theorem excludes uniformity.

8 Reproducible calculations and verification

The companion archive contains this editable L^AT_EX source and PDF, mathematical proof notes, portable Python checks, a dependency list, a SHA-256 manifest, and a Bash replay entry point. It does not bundle third-party full papers. The computations are ordinary exact and symbolic checks; no formal proof-assistant certification is claimed.

The main coefficient calculation uses the cumulant recurrence (3.7). An independent calculation instead forms the truncated raw moment series, applies the finite logarithm series, and then exponentiates in rational polynomial arithmetic. This recovers all four displayed central polynomials and the diagonal coefficients through $n^{-7/2}$, including the zero even coefficients.

A separate dynamic program counts words directly. Its state consists of remaining letter multiplicities and the length of the greedy matched prefix. At each step it chooses a letter with positive remaining count, decrements that count, and advances the prefix exactly when the chosen letter is the next required symbol. It counts distinct words, not copy-labelled permutations. It agrees with (2.2) for all 20 pairs $m = 1, \dots, 4$, $k = 1, \dots, 5$, and independently reproduces Table 2.

Table 2: Exact diagonal counts with the specified offset

n	$a(n)$
0	1
1	1
2	5
3	1306
4	46922017
5	449363984934526

Central numerical checks use the exact integer k and the resulting $x = (k - m)/\sqrt{m}$, rather than a target real x before rounding. Rare-tail checks likewise use the exact k/m . The replay also tests symbolic identities by assertion, so a coefficient mismatch fails the run rather than merely printing a different result. The recorded outputs are reproducibility diagnostics, not numerical proofs of the uniform remainder estimates.

Running the archive. Extract the archive, install the versions listed in `requirements.txt` if needed, and run `bash replay.sh` from the extracted directory. The entry point checks the input manifest, executes every included check, and writes fresh results and logs. Use `bash replay.sh`

--with-pdf to compile the article as well, requiring `pdflatex` and the standard $\text{\LaTeX}$ packages declared in the source. No script relies on executable permission bits, an absolute workspace path, a network call, or a bundled external paper.

[write, 2 October 2026] The archive is not shipped as such. Its scripts are in `code/` and its recorded checks in `data/`, all with the prefix `13-central-`; its proof notes, `SOURCES.md` and `VALIDATION.md` are `13-central-*.md` at the report root. The SHA-256 manifest that `replay.sh` checks first, the delivered PDF and the delivery README are not shipped. The archive survives in the arrival commit `096ee7b87`; the report’s README says how to rerun the replay from it. On Windows Python 3.14 the replay’s final float comparison fails at a relative difference of 2.6×10^{-12} after every exact and symbolic assertion has passed; the README explains why this is a floating-point difference, not a mathematical one.

9 Source audit and bounded novelty

The public sources consulted are the official OEIS entries and the primary papers in the bibliography. The full 2023 Clifton et al. paper and the full El Maazouz–Pitman arXiv paper were inspected. The original 1981 Horton–Kurn publication was not obtained in full; exact enumeration is attributed via its reproduction in Theorem 2.1 of Clifton et al. No claim about absence of asymptotics in the unseen original is made.

The El Maazouz–Pitman paper explicitly supplies the independent Beta construction and the Gaussian theorem. Neither an explicit central Edgeworth correction nor an inverse-threshold expansion was located in the inspected text. Searches combined the sequence identifiers and complete or continuously increasing subsequence terminology with renewal, normal, Edgeworth, asymptotic, and later dates. Such searches cannot exclude an existing general theorem that implies the expansions after specialization. The bounded claim is therefore the explicit family-level formulas and their controlled proofs, with no assertion of a new general probabilistic mechanism.

A read-only overlap check of the ProveIt report repository was pinned to commit `4b874cea0012c51a6841ad9c58f6e20fa57c71da`.

It used a nontruncated SetTheory subtree, the exact OEIS-asymptotics report directory, all 92 Markdown or $\text{\LaTeX}$ files in that directory scope, the full report manifest, and the relevant text members of six incoming archives. Searches found no matching sequence identifiers or Beta-renewal titles in that scoped material. This is a scoped content check, not a claim to have downloaded every repository file. The bibliographic and source-hash record accompanying this report records what was actually inspected.

[write, 2 October 2026] The record is shipped as `13-central-SOURCES.md` and `data/13-central-source-hashes.json`. Since batch 77 this report itself is in the repository, at `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/a047909-beta-renewal-subsequences/`. The pin `4b874cea0` is kept as provenance. At the placement (`4f11bc9c0`) no other repository report mentioned A047909, A268485, El Maazouz–Pitman or Beta renewal.

10 Further research directions

The results leave several concrete questions open within this project.

1. **Uniform matching across the transition.** Construct a single error-function saddle formula valid as $\rho \rightarrow 1$, then prove that its central re-expansion gives the P_j and its separated-ratio re-expansion gives the c_j . The singular coefficients in (7.2) show why substituting $\rho = 1 + x/\sqrt{m}$ into a fixed-ratio remainder is insufficient. A useful result would state a precise shared domain and a relative or absolute error appropriate to each tail.

[write, 2 October 2026] Answered by Part II at every fixed order on every fixed compact positive ratio range. The single formula is the kernel representation $e^A B_R^\varepsilon(u)$ of Theorem 13.1; the shared domain is $a \leq k/m \leq b$; the error is relative for both tails, (13.4). Section 16 proves that its central and separated re-expansions give the P_j and the c_j , explicitly for P_1 and c_1 . Ratios tending to 0 or ∞ and growing orders are not covered.

2. **Moderate deviations.** Determine rigorous growing windows for $x = (k - m)/\sqrt{m}$, with error bounds retaining relative accuracy in a small Gaussian tail. The present fixed- C theorem cannot simply be extended by putting $C = C_m \rightarrow \infty$. One must track the polynomial coefficient growth, the Fourier cutoffs, and the transition from an absolute CDF error to a relative tail error. The regime $\rho - 1 \rightarrow 0$ while $\sqrt{m}|\rho - 1| \rightarrow \infty$ is the natural bridge.

[write, 2 October 2026] Answered in part by Part II. Its Theorem 13.1 covers every sequence with $\rho \rightarrow 1$ and $\sqrt{m}|\rho - 1| \rightarrow \infty$ inside a fixed compact ratio range, with relative accuracy in the small tail. That accuracy is for the unexpanded kernel formula. Part II states that it does not justify extending the bounded-window polynomial expansion (3.1) by substituting a growing normal coordinate. Growing windows for the P_j -expansion itself therefore remain open.

3. **Effective constants and certified rounding.** Extract numerical constants and explicit lower bounds on m from the moment, annulus, and Fourier-tail estimates. Such constants would turn (5.2) into a directly usable certified threshold algorithm, including a principled switch to exact computation near integer boundaries. The present big- O constants are existential and depend on fixed compact sets and fixed truncation order.

[write, 2 October 2026] Still open after Part II, whose constants are also existential (its open question 2 in Section 19).

4. **Stable large-parameter computation.** The exact simplex formula is excellent for audit-sized parameters but has large intermediate alternating integers. Develop a numerically stable renewal convolution or saddle-based method with rigorous interval errors for larger m, k , and compare its cost with exact coefficient convolution and combinatorial dynamic programming.

[write, 2 October 2026] Still open. Part II adds a related question, stable evaluation of its half-line kernels (Section 19, question 1).

5. **High order and truncation.** Study growth of the coefficient families, possible useful truncation rules, and singular behavior as the ratio approaches 0 or 1. Nothing proved here provides growing-order estimates, an optimally truncated transseries, Stokes phenomena, or exponentially improved remainders. Any such extension would need additional analytic work.

[write, 2 October 2026] Still open after Part II (Section 19, question 3), which also leaves ratios tending to 0 or ∞ open.

Conclusion. The established renewal representation makes a controlled quantitative study possible. The central coefficients, diagonal parity, inverses, and separated-ratio tails form two complementary finite-order theories. The remaining challenge is to join their ranges with explicit, transition-uniform and practically effective error estimates.

[write, 2 October 2026] Part II joins the two ranges with transition-uniform error estimates. Its constants are not explicit and its estimates are not practically effective, so that part of the challenge remains.

Part II

Transition uniform Beta renewal asymptotics for complete increasing subsequences

Batch-77 manuscript 14, archive `beta-renewal-uniform-reproducibility.zip` (delivered as `beta-renewal-uniform-addendum.tex`, 12 pages), dated 2 October 2026. Delivered author line: “Research addendum for OEIS A047909 and A268485”. No repository pin; its `source-hashes.json` pins manuscript 13 (Part I) by SHA-256. Printed as delivered, with `[write]` notes; its Section k is Section $k + 10$ here.

Abstract of this Part (as delivered). The central and separated rare-tail expansions for complete increasing subsequences can be joined at every fixed algebraic order. We retain nonsingular Gaussian–Hermite half-line kernels in the classical exponential-tilting and Edgeworth construction and prove a weighted density remainder, uniform over the relevant Beta-renewal triangular family. The resulting approximation has relative error $O(m^{-(R+1)/2})$ for both success and failure, uniformly on every fixed compact positive range of k/m , including moderate deviations and the exact zero-saddle point $k = m + 1$. The first central and separated coefficients are recovered explicitly, and the finite-order matching mechanism is established. A separated-regime corollary gives both real Lambert- W branches for exponentially small target tails, their logarithmic corrections, and qualified integer-threshold brackets. The constants are not effective; endpoint-ratio uniformity, growing truncation order, numerical certification, and general-method novelty are not asserted.

11 What the addendum establishes

Let $H(m, k)$ count words with exactly m copies of each of $1, \dots, k$ that contain the subsequence $12 \cdots k$, and put

$$W(m, k) = \frac{(mk)!}{(m!)^k}, \quad p_m(k) = \frac{H(m, k)}{W(m, k)}. \quad (11.1)$$

The array $H(m, k)$ is OEIS A047909, with multiplicity m as its row index and alphabet size k as its column index [1]. Its diagonal $H(n, n)$ is A268485 for $n \geq 1$; the latter also has $a(0) = 1$ [2].

`[write, 2 October 2026]` (11.1) is Part I’s (1.1). The renewal representation (11.2) below is Part I’s Proposition 2.1 for the scaled increments $Y_{m,i} = mB_i$, so that $\sum B_i \leq 1$ becomes $S_k \leq m$; the density (11.3) is Part I’s (3.11).

The established renewal representation is

$$p_m(k) = \mathbb{P}(S_k \leq m), \quad S_k = Y_{m,1} + \cdots + Y_{m,k}, \quad Y_{m,i} \stackrel{\text{iid}}{\sim} m \text{Beta}(1, m), \quad (11.2)$$

where

$$f_m(y) = (1 - y/m)^{m-1} \mathbf{1}_{(0,m)}(y), \quad \mathbb{E}Y_m = \frac{m}{m+1}. \quad (11.3)$$

The renewal identity and Gaussian limit are prior results of El Maazouz and Pitman [5]; the counting background includes Horton and Kurn [3] and Clifton et al. [4].

The companion report [6] proves central and separated-ratio expansions independently and leaves their transition-uniform joining as an open question. This addendum closes that question on every fixed compact positive ratio range. It is a separate document and does not replace the companion report or alter its results.

[**write**, 2 October 2026] The companion report [6] is Part I. Its open question is the first of its further research directions (Section 10, “Uniform matching across the transition”), whose [**write**] note points back here.

The theorem also covers moderate-deviation sequences that remain in such a ratio range; it does not justify extending a bounded-window central polynomial expansion merely by substituting a growing normal coordinate.

The underlying method is classical. Daniels [7] discusses the Esscher–Cramér approach of tilting, recentering, and integrating an Edgeworth expansion. Related classical constructions include Lugannani and Rice [8], Robinson [9], and the uniformity literature represented by Jensen [10]. The work here is the family-specific verification of a uniform weighted density bound and its conversion to a relative error for both tails, with explicit coefficient matching. Neither a new general saddlepoint formula nor priority for this family-specific theorem is claimed. The representation below is not literally the conventional Lugannani–Rice formula.

12 Exact saddles and nonsingular kernels

Fix $0 < a < b < \infty$ and an integer $R \geq 0$. Unless otherwise stated, all estimates are uniform in integers m, k with $a \leq \rho = k/m \leq b$, as $m \rightarrow \infty$. Constants may depend on a, b, R , and every truncation order is fixed before taking a limit. Define

$$\begin{aligned} M_m(t) &= \mathbb{E}e^{tY_m}, & K_m(t) &= \log M_m(t), & K'_m(t) &= \rho^{-1}, \\ \sigma^2 &= K''_m(t), & \lambda_r &= \frac{K_m^{(r)}(t)}{\sigma^r}, & \eta &= k^{-1/2}, \\ u &= |t|\sigma\sqrt{k}, & A &= kK_m(t) - tm. \end{aligned} \tag{12.1}$$

Here t is the exact real saddle, and all cumulants are the exact m -dependent tilted cumulants, not their limiting values.

12.1 Saddle existence and its exact switch

Under any real tilt, K''_m is a positive variance. Its mean $K'_m(t)$ increases continuously from 0 to m as t runs from $-\infty$ to ∞ . Since $1/\rho \in (0, m)$ for all sufficiently large m , the saddle is unique. On compact intervals below 1, finite expansion of the density gives, with any prescribed finite number of derivatives,

$$\begin{aligned} K_m(t) &= -\log(1-t) + m^{-1}B_1(t) + m^{-2}B_2(t) + O(m^{-3}), \\ B_1(t) &= (1-t)^{-1} - (1-t)^{-2}, \\ B_2(t) &= \frac{3}{2}(1-t)^{-2} - 4(1-t)^{-3} + \frac{5}{2}(1-t)^{-4}. \end{aligned} \tag{12.2}$$

To obtain any fixed order, expand

$$(m-1)\log(1-y/m) = -y + \sum_{j \geq 1} m^{-j} \left(\frac{y^j}{j} - \frac{y^{j+1}}{j+1} \right)$$

only to the needed finite order. Split the transform integral at $y = m^\alpha$ with sufficiently small fixed $\alpha > 0$. Taylor remainders on the first part are bounded by a polynomial times an integrable exponential, and common exponential domination controls the second part. Polynomial weights

give the same estimates for fixed transform derivatives. A finite logarithm then gives (12.2) and its higher-order versions; analyticity in $1/m$ is not required.

Uniform convergence of K'_m on fixed bracketing intervals first puts all saddles in a compact interval $J \subset (-\infty, 1)$. Strict convexity and (12.2) then give

$$t = 1 - \rho + \frac{2/\rho - 1}{m} + O(m^{-2}), \quad \sigma^2 = \rho^{-2} + O(m^{-1}). \quad (12.3)$$

[write, 2 October 2026] (12.2) and (12.3) restate Part I's (7.3) and (7.6), with t for Part I's t_m . Part I uses them only on ratio ranges separated from 1; here the saddle interval contains $t = 0$.

Because $K'_m(0) = m/(m+1)$, the exact sign rule is

$$t > 0 \iff k < m+1, \quad t = 0 \iff k = m+1, \quad t < 0 \iff k > m+1. \quad (12.4)$$

In particular, $k = m$ is not the zero-saddle point.

12.2 Coefficient polynomials and kernels

Using a formal variable v , define

$$C_j(v) = [\eta^j] \exp \left\{ \sum_{r=3}^{R+2} \frac{\lambda_r v^r}{r!} \eta^{r-2} \right\}, \quad 0 \leq j \leq R. \quad (12.5)$$

The symbol $[\eta^j]$ means formal coefficient extraction. Thus

$$C_0 = 1, \quad C_1 = \frac{\lambda_3 v^3}{6}, \quad C_2 = \frac{\lambda_4 v^4}{24} + \frac{\lambda_3^2 v^6}{72}.$$

Replace each v^d in C_j by the probabilists' Hermite polynomial $\text{He}_d(z)$ to obtain $E_j(z)$. Define the real half-line kernels

$$\begin{aligned} H_d(u) &= \int_0^\infty e^{-uz} \phi(z) \text{He}_d(z) dz, & u \geq 0, \\ F(u) &= H_0(u) = e^{u^2/2} \Phi(-u), \\ B_R^\varepsilon(u) &= \sum_{j=0}^R \eta^j \sum_d [v^d] C_j(v) \varepsilon^d H_d(u), & \varepsilon \in \{-1, +1\}. \end{aligned} \quad (12.6)$$

Here ϕ and Φ are the standard normal density and distribution function. Each kernel is finite and smooth at $u = 0$, and

$$H_d(u) = \sum_{q=0}^{\lfloor d/2 \rfloor} \frac{(-1)^q d!}{2^q q! (d-2q)!} (-\partial_u)^{d-2q} F(u). \quad (12.7)$$

In particular, with $\phi(0) = 1/\sqrt{2\pi}$,

$$F'(u) = uF(u) - \phi(0), \quad H_3(u) = (u^2 - 1)\phi(0) - u^3 F(u). \quad (12.8)$$

These are nonsingular analytic formulas. Their direct floating-point evaluation can nevertheless suffer cancellation for large u .

13 The uniform relative tail theorem

Define the outward saddle-side tail and reflection sign by

$$(T, \varepsilon) = \begin{cases} (\mathbb{P}(S_k \geq m), +1), & t \geq 0, \\ (\mathbb{P}(S_k \leq m), -1), & t < 0. \end{cases} \quad (13.1)$$

There is no atom at the threshold, so either convention can be used at $t = 0$.

Theorem 13.1 (Uniform relative accuracy for both tails). *For every fixed $R \geq 0$ and $0 < a < b < \infty$,*

$$T = e^A \left\{ B_R^\varepsilon(u) + O\left(\frac{\eta^{R+1}}{1+u}\right) \right\} = e^A B_R^\varepsilon(u) \{1 + O(\eta^{R+1})\}, \quad (13.2)$$

uniformly for $a \leq k/m \leq b$, with no lower bound on $|t|$. The positive leading term $e^A F(u)$ is uniformly comparable to T . Set

$$\hat{p}_m(k) = \begin{cases} 1 - e^A B_R^+(u), & t \geq 0, \\ e^A B_R^-(u), & t < 0. \end{cases} \quad (13.3)$$

Then, for a constant depending only on the fixed range and order,

$$|\hat{p}_m(k) - p_m(k)| \leq C \eta^{R+1} \min\{p_m(k), 1 - p_m(k)\}. \quad (13.4)$$

Consequently both $\hat{p}_m(k)$ and $1 - \hat{p}_m(k)$ have relative error $O(m^{-(R+1)/2})$ for their respective probabilities.

The outward saddle-selected tail need not be literally the smaller probability near the mean: skewness can put it just above $1/2$. The proof uses only a bounded comparison to the smaller probability, not their equality. At the exact switch,

$$A = u = 0, \quad B_R^+(0) + B_R^-(0) = 1. \quad (13.5)$$

Indeed, every nonconstant C_j contains only positive Hermite degrees, and their full-line Gaussian integrals vanish. The two expressions for $\hat{p}_m(k)$ therefore agree exactly at $t = 0$, without subtracting singular terms or taking a pole-cancellation limit.

The result includes any sequence with $\rho \rightarrow 1$ and $\sqrt{m}|\rho - 1| \rightarrow \infty$, provided ρ stays in a fixed compact positive range. It does not cover $\rho \rightarrow 0$ or $\rho \rightarrow \infty$.

14 A uniform weighted density remainder

This section proves the additional analytic estimate needed for the transition. A uniform distribution-function Edgeworth estimate alone would not control relative error in an exponentially small tail.

Under the tilted density $e^{ty} f_m(y)/M_m(t)$, let

$$X = \frac{Y_m - K'_m(t)}{\sigma}, \quad \chi(v) = \mathbb{E}_t e^{ivX}.$$

Let $f_{m,k,t}$ denote the density of $k^{-1/2} \sum_{i=1}^k X_i$ for independent tilted variables.

Lemma 14.1 (Weighted density estimate). *For every fixed $R \geq 0$ and compact interval $J \subset (-\infty, 1)$, for all sufficiently large m, k ,*

$$\sup_{z \in \mathbb{R}} (1 + z^2) \left| f_{m,k,t}(z) - \phi(z) \sum_{j=0}^R \eta^j E_j(z) \right| \leq C_{R,J} \eta^{R+1}, \quad t \in J. \quad (14.1)$$

The density remainder has both L^1 and L^∞ norm $O_{R,J}(\eta^{R+1})$.

14.1 Uniform analytic and Fourier inputs

Choose $d > 0$ with $\sup J \leq 1 - d$. For sufficiently large m ,

$$e^{ty} f_m(y) \leq e^{-(d/2)y}, \quad y \geq 0, \quad t \in J.$$

Integration over a fixed small interval also gives a positive lower bound on the normalizers. Polynomially weighted dominated convergence is uniform in $t \in J$: the tilted densities and their moments converge to those of exponential laws of rate $1 - t$. The tilted variances remain bounded above and away from zero. The centered standardized variables therefore have a uniform exponential moment in a fixed neighborhood of zero, and uniformly bounded absolute moments of every fixed order.

The characteristic functions extend holomorphically to a common strip near the real axis. On one sufficiently small complex disk, $|\chi(v) - 1| < 1/2$ uniformly, so a local analytic logarithm exists there with uniformly bounded fixed-order derivatives. Exact centering and variance normalization give

$$\log \chi(v) = -v^2/2 + O(v^3), \quad |\chi(v)| \leq e^{-cv^2} \quad (|v| \leq \delta)$$

for fixed positive c, δ . No global logarithm is needed. Uniform convergence of the characteristic functions after standardization gives a strict Cramér gap

$$|\chi(v)| \leq q < 1, \quad \delta \leq |v| \leq L, \quad (14.2)$$

for each fixed L . The limiting standardized exponential characteristic function has modulus $(1 + v^2)^{-1/2}$, independent of its original rate.

For sufficiently large m the tilted density is decreasing, since its logarithmic derivative is

$$t - \frac{m-1}{m-y} < 0 \quad (0 < y < m).$$

Its endpoint value is uniformly bounded, and it tends to zero at m . Integration by parts, followed by the uniformly nondegenerate standardization, gives

$$|\chi(v)| \leq C/|v|. \quad (14.3)$$

Choose L large enough that $C/L < 1$. The derivatives χ' and χ'' are bounded on the entire real axis by uniform absolute moment bounds. Their Fourier decay is not assumed.

14.2 Differentiated Fourier estimates

Write

$$\Psi_k(w) = \chi(\eta w)^k, \quad Q_R(w) = e^{-w^2/2} \sum_{j=0}^R \eta^j C_j(iw), \quad D_R(w) = \Psi_k(w) - Q_R(w). \quad (14.4)$$

We prove

$$\|D_R\|_1 + \|D_R'\|_1 \leq C\eta^{R+1}. \quad (14.5)$$

Fix a sufficiently small $\gamma > 0$, for example $\gamma < 1/[12(R+3)]$.

On $|w| \leq k^\gamma$, Taylor expansion of the local logarithm gives

$$k \log \chi(\eta w) = -\frac{w^2}{2} + \sum_{r=3}^{R+2} \frac{\lambda_r(iw)^r}{r!} \eta^{r-2} + \mathcal{R}(w). \quad (14.6)$$

Applying Taylor's theorem separately to the logarithm and its first two derivatives gives

$$|\mathcal{R}^{(\ell)}(w)| \leq C\eta^{R+1}|w|^{R+3-\ell}, \quad \ell = 0, 1, 2. \quad (14.7)$$

To make the finite-order bookkeeping precise, set $G(w) = \sum_{j=1}^R \eta^j a_j(w)$, where a_j has degree $j+2$. Expand e^G through $G^R/R!$ with its integral remainder. Every product omitted from the finite polynomial has total η -degree at least $R+1$. The same is true after at most two w -derivatives. In the integral remainder, each factor of G or one of its derivatives carries at least one power of η . All such terms are bounded by η^{R+1} times a fixed polynomial in $|w|$. The choice of γ keeps G and $\mathcal{R}$ bounded on this range, and a slightly weakened Gaussian absorbs their exponentials. The retained coefficients are precisely $C_j(iw)$. Thus, for a fixed polynomial P_R ,

$$|D_R^{(\ell)}(w)| \leq C\eta^{R+1}P_R(|w|)e^{-c'w^2}, \quad \ell = 0, 1, 2. \quad (14.8)$$

This argument also covers $R = 0$, when the finite correction sum is empty. Integration gives the required order on the low range.

On $k^\gamma < |w| \leq \delta\sqrt{k}$, use the exact derivatives

$$\begin{aligned} \Psi'_k(w) &= \sqrt{k} \chi^{k-1} \chi', \\ \Psi''_k(w) &= (k-1) \chi^{k-2} (\chi')^2 + \chi^{k-1} \chi'', \end{aligned} \quad (14.9)$$

where all factors on the right are evaluated at ηw . The local Gaussian bound and bounded derivatives show that the integrals of Ψ_k and its first two derivatives are at most a polynomial in k times $e^{-c''k^{2\gamma}}$. The corresponding Gaussian-polynomial Q_R terms have negligible integrals too.

On $\delta\sqrt{k} \leq |w| \leq L\sqrt{k}$, the same identities and (14.2) give an integrated bound $Ck^{3/2}q^{k-2}$. On $|w| > L\sqrt{k}$, (14.3) gives

$$Ck^{3/2} \int_L^\infty (C/v)^{k-2} dv = Ck^{3/2} \frac{L}{k-3} (C/L)^{k-2}, \quad k > 3, \quad (14.10)$$

with a harmless change of the outside constant. This also decays exponentially. These estimates control all three derivative orders 0, 1, 2 and prove (14.5).

14.3 Fourier inversion and the two norms

Let r be the density remainder in (14.1). The Hermite convention ensures that the inverse Fourier transform of $(iw)^d e^{-w^2/2}$ is $\phi(z) \text{He}_d(z)$. Hence inversion gives

$$|r(z)| \leq \frac{\|D_R\|_1}{2\pi}, \quad |z^2 r(z)| \leq \frac{\|D_R''\|_1}{2\pi}.$$

The second estimate follows from two integrations by parts: D_R, D_R' vanish at infinity, and the required derivatives are integrable, by the same high-frequency bounds above. Adding the two estimates proves (14.1). Taking the supremum gives the L^∞ estimate; integrating $(1+z^2)^{-1}$ gives the L^1 estimate. This completes the proof of Lemma 14.1.

15 From density control to relative tail control

At the exact saddle, under the tilted law, $S_k = m + \sigma\sqrt{k}Z$, where Z has density $f_{m,k,t}$. The change of measure and reflection $Z = \varepsilon z$ give the exact identity

$$T = e^A \int_0^\infty e^{-uz} f_{m,k,t}(\varepsilon z) dz. \quad (15.1)$$

If r is the density remainder, its integral in this formula is at most $\|r\|_1$ when $0 \leq u \leq 1$, and at most $\|r\|_\infty / u$ when $u \geq 1$. Lemma 14.1 therefore gives the additive remainder in (13.2), uniformly through $u = 0$.

For each fixed d , the same two-norm argument gives $|H_d(u)| \leq C_d/(1+u)$. Moreover,

$$F(u) \asymp (1+u)^{-1}, \quad u \geq 0. \quad (15.2)$$

For the upper bound, use $F \leq 1/2$ and $\phi \leq \phi(0)$. For the lower bound, restrict the defining integral to $0 \leq z \leq 1/(1+u)$, on which $e^{-uz} \geq e^{-1}$ and $\phi(z) \geq \phi(1)$. Since all fixed cumulants are uniformly bounded,

$$B_R^\varepsilon(u) = F(u)\{1 + O(\eta)\},$$

and this bracket is positive for all sufficiently large k . Division proves the relative version of (13.2).

Convexity, together with $K_m(0) = 0$ and $K'_m(t) = m/k$, implies $A \leq 0$. The $R = 0$ case and $F(u) \leq 1/2$ show that $T \leq 1/2 + C\eta$, so $1 - T \geq 1/3$ for all sufficiently large k . Consequently $T/\min(T, 1 - T)$ stays bounded, even if T is slightly larger than $1/2$. The relative estimate for T thus gives (13.4), completing the proof of Theorem 13.1.

16 Recovery of the two earlier regimes

16.1 The first central correction

Set $h = m^{-1/2}$ and $k = m + x\sqrt{m}$, with x in a fixed compact interval. The actual integer k determines x ; no independent rounding of x is implicit. Finite re-expansion gives

$$\begin{aligned} t &= -xh + h^2 + O(h^3), & \sigma &= 1 - xh + O(h^2), \\ t\sigma\sqrt{k} &= -x + (1 + x^2/2)h + O(h^2), & A &= -x^2/2 + (x + x^3/6)h + O(h^2), \\ \eta &= h + O(h^2), & \lambda_3 &= 2 + O(h^2). \end{aligned} \quad (16.1)$$

For fixed $x > 0$, the lower-tail branch applies. Its coefficient of h , after removal of $e^{-x^2/2}$, is

$$\begin{aligned} (x + x^3/6)F(x) - (1 + x^2/2)F'(x) - H_3(x)/3 \\ = \frac{x^2 + 8}{6}\phi(0), \end{aligned}$$

by (12.8). The upper/complement branch for $x < 0$ gives the same result. Thus

$$p_m(k) = \Phi(-x) + m^{-1/2}\phi(x)P_1(x) + O(m^{-1}), \quad P_1(x) = \frac{x^2 + 8}{6}. \quad (16.2)$$

This matching is uniform through the sign change. Away from the shrinking switch region, the displayed branch re-expansions have uniform Taylor errors. On $|x| \leq Ch$ the quantities satisfy $u = O(h)$ and $A = O(h^2)$, and

$$F(u) = \frac{1}{2} - \phi(0)u + O(u^2), \quad H_3(u) = -\phi(0) + O(u^2).$$

Either branch, with the complement relation at zero, gives

$$p_m(k) = \frac{1}{2} + \phi(0)(-x + 4h/3) + O(h^2),$$

which agrees with (16.2). The apparent absolute-value branching therefore introduces no transition defect.

[write, 2 October 2026] (16.2) is Part I's Theorem 3.1 at $R = 1$, with P_1 of (3.2). Here it is derived from Theorem 13.1 by re-expansion, a second route, and is printed as a consistency check, not as a second statement of Part I's theorem.

At $k = m + 1$, the lower-half expression at zero tilt gives

$$p_m(m+1) = \frac{1}{2} + \frac{\lambda_3 \phi(0)}{6\sqrt{k}} + O(k^{-3/2}). \quad (16.3)$$

The order- k^{-1} term vanishes because its Hermite polynomials are even, of positive degree, and have zero half-line Gaussian integral. The use of $R = 2$ justifies the displayed remainder. The first correction is $\phi(0)h/3$, agreeing with (16.2) at $x = h$. At $k = m$, the positive-saddle complement branch instead gives $1/2 + 4\phi(0)h/3 + O(h^2)$.

16.2 The first separated correction

Suppose now that ρ lies in a compact set contained either in $(0, 1)$ or in $(1, \infty)$. Then $|t|$ is bounded away from zero. Set $w = t\sigma$, so $u = |w|/\eta$. Use Theorem 13.1 with $R = 3$, whose relative error is $O(\eta^4)$. Watson expansion of the finitely many kernels gives

$$\begin{aligned} F(u) &= \phi(0)\{u^{-1} - u^{-3} + O(u^{-5})\}, \\ H_3(u) &= \phi(0)\{-3u^{-2} + O(u^{-4})\}, \\ H_4(u) &= \phi(0)\{3u^{-1} + O(u^{-3})\}, \\ H_6(u) &= \phi(0)\{-15u^{-1} + O(u^{-3})\}. \end{aligned} \quad (16.4)$$

The odd-degree kernels in E_3 are $O(u^{-2})$, so their relative contribution is $O(\eta^4)$, not order η^3 . It follows that

$$\begin{aligned} T &= \frac{e^A}{|t|\sqrt{2\pi k\sigma^2}} \{1 + Q_1/k + O(k^{-2})\}, \\ Q_1 &= \frac{\lambda_4}{8} - \frac{5\lambda_3^2}{24} - \frac{\lambda_3}{2t\sigma} - \frac{1}{t^2\sigma^2}. \end{aligned} \quad (16.5)$$

The signed factor t in the skew term is essential on both sides.

Define

$$I(\rho) = 1 - \rho + \rho \log \rho, \quad A_0(\rho) = \frac{e^{1-1/\rho} \sqrt{\rho}}{|\rho - 1| \sqrt{2\pi}}. \quad (16.6)$$

Substituting (12.2) and the saddle expansion into (16.5) gives

$$T = m^{-1/2} A_0(\rho) e^{-mI(\rho)} \{1 + c_1(\rho)/m + O(m^{-2})\}. \quad (16.7)$$

Here $T = 1 - p_m(k)$ below ratio 1 and $T = p_m(k)$ above it. The three contributions to c_1 are, respectively,

$$\frac{2\rho^2 - 4\rho + 1}{2\rho^3}, \quad -\frac{\rho^2 - 3\rho + 1}{\rho^2(\rho - 1)}, \quad \frac{1}{\rho} \left\{ -\frac{1}{12} - \frac{\rho}{(1 - \rho)^2} \right\}, \quad (16.8)$$

from the exponent, the prefactor, and Q_1/k . Their sum is exactly

$$c_1(\rho) = -\frac{\rho^4 + 10\rho^3 - 17\rho^2 + 24\rho - 6}{12\rho^3(\rho - 1)^2}, \quad (16.9)$$

recovering the coefficient of the companion report.

[write, 2 October 2026] The companion report's coefficient is Part I's (7.2). Q_1 in (16.5) is Part I's (7.11), and the three contributions (16.8) are those in Part I's proof of Theorem 7.1. Part I reaches them from its exact-saddle expansion (7.10); here they come from the uniform kernel formula through Watson expansion. This is a consistency check by a second route, not a second statement of Part I's Theorem 7.1.

16.3 Matching at every fixed order

Every monomial contributing to C_j has degree congruent to j modulo 2: if its cumulant factors have orders $r_1, \dots, r_s$, then its η -degree is $\sum(r_i - 2) = j$ and its polynomial degree is $\sum r_i = j + 2s$. An even Hermite kernel has inverse- u powers $u^{-1}, u^{-3}, \dots$; an odd kernel has powers $u^{-2}, u^{-4}, \dots$. After division by the leading η factor, every term therefore has an even total power of η . For any fixed separated correction order J , choose $R = 2J + 1$. Then the relative remainder is $O(\eta^{2J+2})$, and finite kernel and transform re-expansions recover all separated coefficients through m^{-J} . Their identification with the earlier c_j follows from uniqueness of finite asymptotic coefficients.

In the bounded central window, finite re-expansion of the same formula to any requested fixed order has a controlled remainder by Theorem 13.1. Comparison with the companion report's uniform central expansion identifies all its P_j by the same uniqueness argument. The explicit calculations of P_1 and c_1 above independently check normalization, reflection signs, and the exact location of the switch.

[write, 2 October 2026] This section thus gives a second route to the existence of fixed-order central and separated expansions, by re-expansion of Theorem 13.1. It identifies their coefficients with Part I's P_j and c_j by uniqueness, which presupposes Part I's expansions; only P_1 and c_1 are computed explicitly here. Part I's own proofs and coefficient algorithms (Theorems 3.1 and 7.1) are kept. For the diagonal, $k = m$ is not the zero-saddle point ((12.4)); Part I's Corollary 4.1 is the case $x = 0$ of its central expansion.

17 Separated rare tail thresholds

This corollary uses the separated expansion and is not a claim of transition-uniform inversion. Let the target probability be $e^{-m\ell}$. For rare success let ℓ vary in a fixed compact subset of $(0, \infty)$; for rare failure let it vary in a fixed compact subset of $(0, 1)$. In particular, these ranges exclude $\ell = 0$ and, for failure, the endpoint $\ell = 1$.

17.1 The two Lambert branches

The two solutions of $I(\rho_0) = \ell$ are

$$\rho_0 = \begin{cases} \exp\{1 + W_0((\ell - 1)/e)\} > 1, & \text{rare success,} \\ \exp\{1 + W_{-1}((\ell - 1)/e)\} < 1, & \text{rare failure.} \end{cases} \quad (17.1)$$

Indeed, $w = \log \rho_0 - 1$ gives $we^w = (\ell - 1)/e$. The principal real branch has $w > -1$ in the success range; the lower real branch has $w < -1$ in the failure range. Monotonicity of I on each side of 1 proves uniqueness. The compact restrictions keep ρ_0 in a compact positive set separated from 1.

[write, 2 October 2026] (17.1) is an exact Lambert- W solution of the rate equation $I(\rho_0) = \ell$. The corrections d_0, d_1 of Corollary 17.1 are an instance of the perturbed inversion `p0:thm:perturbed-inversion` of the repository's transseries volume, and the brackets (17.7) are of the first-crossing kind of its staircase recovery `p0:thm:staircase`. No novelty is claimed for the inversion method. The separated rate here is Part I's $I(\rho)$, and the amplitude $A_0(\rho)$ is the prefactor of Part I's (7.1) without $m^{-1/2}$.

Corollary 17.1 (Smooth carrier inversion). *Take a sufficiently high finite separated expansion and continue its formula smoothly to real k using $\rho = k/m$. This defines a carrier, not a canonical real- k extension of the exact integer probability. Its locally unique crossing $K_m(\ell)$ satisfies, uniformly on the stated compact ℓ ranges,*

$$K_m(\ell) = m\rho_0 + \frac{L(\rho_0) - \frac{1}{2} \log m}{\log \rho_0} + O((\log m)^2/m), \quad L = \log A_0. \quad (17.2)$$

More explicitly, put

$$\begin{aligned} g &= \log \rho_0, & d_0 &= \frac{L(\rho_0) - \frac{1}{2} \log m}{g}, \\ L'(\rho) &= \frac{1}{\rho^2} + \frac{1}{2\rho} - \frac{1}{\rho - 1}, \\ d_1 &= \frac{-d_0^2/(2\rho_0) + L'(\rho_0)d_0 + c_1(\rho_0)}{g}. \end{aligned} \tag{17.3}$$

Then

$$K_m(\ell) = m\rho_0 + d_0 + d_1/m + O((\log m)^3/m^2). \tag{17.4}$$

Proof. The logarithm of the carrier is $-mI(\rho) - \frac{1}{2} \log m + L(\rho) + c_1(\rho)/m + O(m^{-2})$. Write $\rho = \rho_0 + d/m$. Cancellation of the target exponent gives

$$\begin{aligned} 0 &= -gd - \frac{1}{2} \log m + L(\rho_0) \\ &\quad + m^{-1}\{-d^2/(2\rho_0) + L'(\rho_0)d + c_1(\rho_0)\} + O((1 + |d|^3)/m^2). \end{aligned} \tag{17.5}$$

First $d = O(\log m)$. The choices d_0 and d_1 cancel the next two orders. The derivative with respect to d is $-g + O(\log m/m)$, uniformly separated from zero. The residual is therefore also a root-error bound, giving (17.4) and local uniqueness. Continuing this finite implicit expansion yields any fixed further order, with polynomials in $\log m$; powers of $\log \rho_0$ in the denominators explain the excluded transition. $\square$

17.2 What the inverse says about integer crossings

For the exact probabilities, define

$$q_s = \min\{k \geq 1 : p_m(k) \leq e^{-m\ell}\}, \quad q_f = \min\{k \geq 1 : 1 - p_m(k) \geq e^{-m\ell}\}. \tag{17.6}$$

Positivity of the increments makes $p_m(k)$ nonincreasing in k , and the relevant separated crossings are well defined. Set

$$\kappa = m\rho_0 + d_0 + d_1/m, \quad \Delta = C(1 + \log m)^3/m^2.$$

For a sufficiently large uniform constant C and sufficiently large m , the controlled carrier/probability error and the nonzero local logarithmic slope give, for the corresponding branch,

$$\lceil \kappa - \Delta \rceil \leq q_s \leq \lceil \kappa + \Delta \rceil \quad \text{or} \quad \lceil \kappa - \Delta \rceil \leq q_f \leq \lceil \kappa + \Delta \rceil. \tag{17.7}$$

The constant absorbs both the smooth root error and the probability approximation error. Monotonicity turns their local sign bounds into bounds for the first integer crossing.

The constant and the lower cutoff for m are existential here, so (17.7) is not a finite- m numerical certificate. Even once effective bounds are available, a bracket containing two integers requires exact or certified evaluation to decide between them. An unconditional ceiling of the displayed truncated smooth expansion is not justified. The integer threshold itself does not have a vanishing-error smooth expansion.

18 Reproducibility and source scope

The accompanying archive contains this L^AT_EX source, the PDF, portable build and replay scripts, pinned Python dependencies, recorded finite checks, a file-integrity manifest, and a bounded source-scope record. It contains no third-party full-text articles. Running `bash replay.sh` checks the manifest, reruns the symbolic and numerical calculations in a fresh output directory, and compares their outputs to the recorded checks. `bash replay.sh -with-pdf` also rebuilds the article when a suitable L^AT_EX installation is available. Extraction with Python’s `zipfile` is supported; executable mode bits are not required.

Two independent forms of checking accompany the proof. Symbolic calculations rederive the first saddle displacement, the variance, exponent, and prefactor corrections, the H_3 identity, and the P_1 and c_1 identities. Degree-parity checks through order 10 are finite diagnostics; the all-order parity proof is the argument in Section 16. Numerical diagnostics compare the $R = 0, 1, 3$ formulas with exact rational probabilities at 16 pairs (m, k) , including both tails and the exact switch. They evaluate moments and kernels by high-precision, non-interval quadrature. They support implementation checks and do not prove the theorem or supply certified error constants.

On that finite sample the largest observed values of $k^{(R+1)/2}|\hat{p}-p|/\min(p, 1-p)$ are approximately 0.3072, 0.08194, and 0.03466 for $R = 0, 1, 3$, respectively. These are sample maxima, not proven uniform constants. The threshold checks verify rate-branch identities and cancellation of the displayed inverse coefficients; no exact-rounding certification is inferred.

[write, 2 October 2026] The archive is not shipped as such. Its scripts are in `code/` and its recorded checks in `data/`, all with the prefix `14-uniform-`; its `SOURCES.md` and `VALIDATION.md` are `14-uniform-*.md` at the report root. Its manifest, delivered PDF, build script and delivery README are not shipped; the archive survives in the arrival commit `096ee7b87`, and the report’s README says how to rerun the replay from it. The companion report’s primary-source record mentioned below is `13-central-SOURCES.md`.

Literature boundaries. The general-method attribution uses the inspected first-page introduction and publisher abstract of Daniels [7]. Pages 38–48 of that article were not inspected, and no exact theorem from those pages is invoked. The original publisher text of Lugannani and Rice [8], especially pp. 478–483, documents tilted associated densities and classical uniform tail-series constructions, but is not used as a substitute for the triangular family proof here. Robinson [9] was consulted at the abstract and indexed-publisher-text level; Jensen [10] at the abstract level. Their precise formulas and hypotheses are not claimed to imply Lemma 14.1. For the counting literature, the companion report’s primary-source record supplies the earlier audit scope. The full Horton–Kurn article was not inspected; its enumeration was consulted through Clifton et al. This is a bounded attribution review, not an exhaustive novelty search.

19 What remains open

The analytic transition question is resolved within the stated fixed order and compact ratio scope. The following separate questions remain useful.

1. **Stable evaluation of the kernels.** Develop an implementation that evaluates the scaled normal tail and Gaussian–Hermite half-line kernels reliably for both small and large u . Direct derivative-polynomial identities can cancel badly at large u . Switching rules, recurrences, quadrature, or asymptotic kernel evaluation need quantitative error analysis.
2. **Effective constants and certified thresholds.** Make the moment bounds, Cramér gap, Fourier cutoffs, and density remainder constants explicit. Combine these with validated saddle

and kernel evaluation to turn the existential integer brackets into usable certificates, with exact computation near an integer boundary.

3. **Ratio endpoints and order growth.** The present theorem keeps k/m in a fixed compact subset of $(0, \infty)$ and fixes R . Ratios tending to 0 or infinity, growing orders, coefficient growth, and optimal truncation require new estimates. No convergent asymptotic series, exponentially improved remainder, or Stokes analysis is implied.
4. **Uniform inverse regimes.** The Lambert inverses deliberately avoid $\ell = 0$, where $\log \rho_0$ vanishes, and the failure endpoint $\ell = 1$. A unified inversion with explicit error control as target tails move between central and separated scales is a further problem, even though the forward probabilities now have a uniform representation.

Conclusion. Keeping the half-line kernels unexpanded removes the transition singularities. Uniform control of the tilted density in both norms then supplies the same relative order for both tails, while finite re-expansion recovers the earlier central and rare-tail theories. This is a mathematical uniformity result; a stable, effective, certified numerical method remains to be built.

References

- [1] OEIS Foundation Inc., *A047909: Array read by antidiagonals upwards*, <https://oeis.org/A047909>, inspected 2 October 2026. Multiplicity is the row index and alphabet size is the column index.
- [2] OEIS Foundation Inc., *A268485: Number of sequences with n copies each of $1, 2, \dots, n$ and longest increasing subsequence of length n* , <https://oeis.org/A268485>, inspected 2 October 2026.
- [3] J. D. Horton and A. Kurn, *Counting sequences with complete increasing subsequences*, *Congressus Numerantium* **33** (1981), 75–80. Original not inspected in full; enumeration consulted through [4, Theorem 2.1].
- [4] A. Clifton, B. Deb, Y. Huang, S. Spiro and S. Yoo, *Continuously increasing subsequences of random multiset permutations*, *European Journal of Combinatorics* **110** (2023), 103708. doi:10.1016/j.ejc.2023.103708. Preprint: <https://arxiv.org/abs/2110.10315>.
- [5] Y. El Maazouz and J. Pitman, *The Bernoulli clock: probabilistic and combinatorial interpretations of the Bernoulli polynomials by circular convolution*, *Combinatorics, Probability and Computing* **33** (2024), 210–237. Published online 16 November 2023. doi:10.1017/S0963548323000421. Preprint: <https://arxiv.org/abs/2210.02027>.
- [6] *Quantitative Beta renewal asymptotics for complete increasing subsequences*, companion research report for OEIS A047909 and A268485, 2 October 2026. Central and separated-regime results; not a published journal article. [write] Printed as Part I of this report.
- [7] H. E. Daniels, *Tail Probability Approximations*, *International Statistical Review* **55**(1) (1987), 37–48. doi:10.2307/1403269.
- [8] R. Lugannani and S. Rice, *Saddle point approximation for the distribution of the sum of independent random variables*, *Advances in Applied Probability* **12**(2) (1980), 475–490. doi:10.2307/1426607.
- [9] J. Robinson, *Saddlepoint Approximations for Permutation Tests and Confidence Intervals*, *Journal of the Royal Statistical Society, Series B* **44**(1) (1982), 91–101. doi:10.1111/j.2517-6161.1982.tb01191.x.
- [10] J. L. Jensen, *Uniform saddlepoint approximations*, *Advances in Applied Probability* **20**(3) (1988), 622–634. doi:10.2307/1427038.